

Next GM V2 — Game Mechanics Audit & Review Guide

This document maps **every mechanical layer** in the current prototype: what is calculated, where, with what constants, and how systems chain together. Use it as the shared reference for our balance/strengthening review.

Architecture: The Playable Loop as a Math Pipeline

Setup and Draft: NewGameSetupScreen → simulateOpeningDraft → createNewGame

Weekly Loop: Booking Card → runShow (scoring.ts) → Results + Fallout → Week Review → advanceGameWeek → (repeat)

Parallel CPU World: runShow triggers generateCpuWeeklyResults; advanceGameWeek triggers advanceCpuMarket

Core design principle (enforced in code): Pre-show surfaces show **context and warnings only**. Scores, winners, Open Challenge opponents, injuries, morale changes, title changes, finance P/L, and social posts resolve **only after Run Show** via deterministic formulas + hash-based variance.

Determinism: Same save + same card → same outcome. RNG is pseudo-random via hashString in src/game/scoring.ts (mod 1_000_000_007), seeded with season/week/segment/wrestler IDs.

1. Draft & Career Start

Key files

- src/game/top200DraftPool.ts — 120-wrestler pool, Madden-like stat calibration
- src/game/openingDraft.ts — parallel CPU draft simulation
- src/game/draftOrder.ts — snake/linear/random/lottery order
- src/game/seed.ts — createNewGame, starting money, default titles/rivalries
- src/game/financeCatalog.ts — per-wrestler draftValueUsd
- data/finance/roster_draft_and_contract_values_top_200_no_bonus_full_season_2026-05-16.csv

Stat seeding (draft pool)

Each wrestler gets stats from **role-tier bands** blended with draft rank percentile:

Stat	Blend weights (statPercentile vs draftPercentile)
Popularity	35% / 65%
Momentum	75% / 25%
Morale	85% / 15%
Ring/Promo	65% / 35%

Hard caps: pop ≤94, mom ≤88, ring/promo ≤92, morale ≤72. Archetype nudges (e.g. Technician +5 ring).

Display OVR (UI only): $\text{pop} \times 0.24 + \text{mom} \times 0.22 + \text{ring} \times 0.18 + \text{promo} \times 0.16 + \text{morale} \times 0.12 + (100 - \text{fatigue}) \times 0.08$, clamped 40–99.

Draft finance

- **Starting budgets:** \$1M / \$2M (default) / \$4M / Unlimited (\$999,999,999)
- **Draft cost:** CSV draftValueUsd per wrestler (~\$30k–\$260k by role tier)
- **Bundle discount:** 20% off tag/faction packages (×0.8)
- **Opening money:** startingBudget – rosterDraftSpend + bundleDiscount
- **Player affordability gate:** pick blocked if draftCost > projectedReserve (Money Based only)
- **CPU rival budget:** \$1,800,000; drafts until no affordable pick remains

CPU draft scoring (scoreCpuDraftCandidate)

```
talentScore = rankScore + pop×0.46 + max(ring,promo)×0.34 + sourceBrandBonus(10)
valueScore = max(0, 180000 – draftCost) / 4500
total = talent×talentWeight + style×styleWeight + divisionNeed×needWeight
        + value×valueWeight + roundNoise×noiseWeight
```

Difficulty scales weights (Legendary: +8% talent, +36% roster need, –42% noise). Tie-break: lower draftRank.

Career bootstrap (createNewGame)

- Player roster + disjoint CPU rosters
- Default championships from starting roster
- 3 opening rivalries from top star pairs
- Market board seeded; rival brands get CPU depth (titles, rivalries, contracts)

Review notes

- **Difficulty does not currently map to starting budget** in setup UI — only CPU/market/player-pressure tuning
- Draft is fully re-simulated on every player pick change (deterministic rewind)
- Minimum roster: **12** to start; recommended target **16** (advisory only)

2. Show Simulation (Core Engine)

Single entry point: runShow in src/game/scoring.ts

Pipeline order

1. Validate segments → resolve runtime durations (120-min broadcast window)
2. Per segment: Open Challenge resolution → score → momentum/fatigue → title → rivalry
3. Aggregate show score + grade
4. updateWrestlerPressure (morale + TV tracking for full roster)
5. evaluateInjuries (deterministic roll)
6. Title stat fallout (momentum/popularity, not morale)
7. Social inbox morale resolution
8. generateFinanceReport → update money
9. generateCpuWeeklyResults → rival standings
10. generateSocialPosts

Segment scoring (scoreSegment)

Per participant, weighted stat sum averaged across participants:

Segment type	ring	promo	pop	mom	morale	fatigue(sub)
Match	0.44	—	0.25	0.19	0.10	0.15
Promo	—	0.43	0.27	0.20	0.10	0.13
Backstage Angle	—	0.30	0.18	0.27	0.11	0.10
Contract Signing	—	0.31	0.25	0.16	0.11	0.10
Open Challenge	0.38	—	0.24	0.20	0.10	0.17

Bonuses stacked on raw score:

- Chemistry: Match +3, Promo +2, others +1 (multi-participant only)
- Rivalry context: +3 (or +4 if title stakes)
- Championship context: +2 to +3 by segment type
- Stipulation scoreBonus: +1 to +4 (stipulationCatalog.ts)
- PLE: +4
- Overrun segment penalty: minor 4+ctx, moderate 8+ctx, major 14+ctx

Show score: $\text{round}(\text{avg}(\text{segmentScores})) - \text{overrunShowPenalty}$ (minor −1, moderate −4, major −8)

Grades: A≥90, B≥80, C≥70, D≥60, F otherwise

Match winners (getWinnerScore) — no randomness

$$\text{pop} \times 0.30 + \text{mom} \times 0.25 + \text{ring} \times 0.35 + \text{morale} \times 0.15 - \text{fatigue} \times 0.18$$

Player can override via segment.winnerId. Tag (M020): team scores summed; tie → Team A.

Open Challenge (resolveOpenChallenge)

- Opponent picked at show-run: lowest hash(season-week-segmentId-index-wrestlerId)
- Eligibility: same division group, available, not protected rest; prefer unbooked tonight
- **Moment modifier:** readiness gap (±2/−3), fatigue drag (−2 at ≥50), momentum spark (+2 at ≥65)
- No title changes; no contest → score 0, fatigue +1

Momentum gain (getSegmentMomentumGain)

Segment score	Gain
≥95	+8
≥90	+7
≥85	+6
≥70	+4
≥55	+2
<55	+1

PLE: +2 extra. Applied to **every participant**.

Fatigue gain (getSegmentFatigueGain)

```
max(typeFloor, fatigueBase + duration×fatiguePerMinute + priorCardUses×2 +
minorInjuryLoad)
```

Type floors: Match 8, OC 7, Backstage 3, Promo/Contract 2. PLE +1. Stipulation fatigue bonus. Same wrestler twice on card: +2 per prior use.

Weekly recovery: −3 fatigue on Advance Week (advanceWeek.ts)

3. Rivalry System

Constants: RIVALRY_BALANCE in scoring.ts

Heat movement per segment (resolveRivalrySegment)

Score	Heat Δ
≥95	+12
≥90	+9
≥75	+6
≥60	+3
<60	−6

Additions: Backstage +1, Contract +2, stipulation heat bonus, PLE +5, PLE payoff (≥85) +3, title stakes +2

Freshness (story repetition tax)

- Normal beat: −3 (elite ≥95: −3; cold weak beat: up to −7)
- Repeated beat same night: −11 to −15 (punishes spamming same feud on one card)

Crowd-sparked rivalry (auto-create)

Match, 2 participants, no existing rivalry, score ≥85, same division:

```
heat = clamp(score×0.55 + avg(pop,mom)×0.45)
freshness = PLE/title ? 86 : 82
```

Status thresholds (getRivalryStatus)

- Stale: freshness ≤25
- Cooling: heat <45 or freshness <45
- Rising: heat ≥70 and freshness ≥45

Advance-week decay (no beat that week)

```
heat -= 4 (+4 if stale)
freshness -= 3 (+4 if stale)
```

4. Championships & Title Fallout

Title changes: Match segments only (resolveTitleMatch). Non-match title context adds score bonus but cannot change belt.

Post-title stat fallout (titleStatFallout.ts):

Event	Momentum	Popularity
Title win	+8	+4
Title loss	−4	−2
Defense (champ)	+3	+1
Challenger loss	−1	0

Scaled $\times 1.25$ on PLE, $\times 0.85\text{--}1.15$ by prestige. High-quality match bonuses at score $\geq 85/90$.

5. Roster Pressure, Morale & Injuries

There is no separate "sentiment" stat — locker-room sentiment = **morale** (0–100). Audience mood = Social/IWC posts (retrospective).

Morale fallout (updateWrestlerPressure) — post-show, all wrestlers

Condition	Morale Δ
Booked tonight	+2
Return after ≥ 3 weeks off	+4 bonus
Protected rest (social inbox)	+2, no underuse penalty
Booked while fatigued (≥ 50) or overused (≥ 3 consecutive weeks)	−4 (or −6 if fatigue ≥ 76)
Not booked, ≥ 3 weeks off TV	−3
Social: TV request fulfilled/broken	+3 / −4

Penalties scaled by difficulty moralePenaltyMultiplier (Easy 0.75 $\rightarrow$ Legendary 1.30).

TV time tracking

- Booked: appearancesThisSeason++, lastBookedWeek = currentWeek, streak +1
- Not booked: consecutiveWeeksBooked = 0
- Open Challenge resolved opponents count as booked

Momentum decay (Advance Week)

- Unbooked: **−4** momentum
- Booked with momentum ≥ 92 : **−2** (elite decay)

Injury system (evaluateInjuries)

Risk score:

```

fatigueLoad = highestFatigue×0.7 + (≥85 ? 8 : ≥70 ? 4 : 0)
repeatLoad = consecutiveWeeks×4
physicalLoad = physicalSegments×12 + (≥2 physical ? 14 : 0)
minorInjuryLoad = 12 if already minor
stageLoad = 2 on PLE
+ difficulty.injuryRiskModifier (−5 to +7)

```

Roll: risk + hash%20

- ≥97 → **major** (3–8 weeks, blocks booking)
- ≥92 → **minor** (1–3 weeks, can still work)
- else healthy

Pressure labels (UI advisory, rosterContextReads.ts)

Overused, Underused, Morale Risk (≤45), Injury Risk (fatigue ≥75), Protected Star, Unavailable (major injury)

6. Finance

Key file: src/game/finance.ts — model show-production-finance-v3

Weekly show report (Run Show only)

Revenue drivers: attendance (from show score, average draw, segments, title matches), tickets, merch, media rights.

Draw inputs:

```

averageDraw = avg(pop×0.5 + mom×0.25 + max(ring,promo)×0.25)
merchAppeal = avg(pop×0.45 + mom×0.35 + promo×0.2)
scoreFactor = clamp((totalScore - 50) / 50, 0, 1)

```

Expenses: base production (\$65k TV / \$240k PLE) + segment catalog costs + stipulation costs + booked-finish fee (\$10k/match) + overrun costs.

Important: Weekly reports do **not** include payroll or market transactions. Contracts are **prepaid upfront** (market.ts — getActivePlayerPayroll() returns 0).

Market cash flow (immediate on action)

- Sign/renew: money −= dueNow (weeklyAsk × weeks, with volatility hash)
- Release: penalty = 0 for prepaid model
- Trade fee: max(0, |valueDelta| × 1400) if accepted

Office mandate (evaluateOfficeMandate) — Advance Week

Trust/reputation deltas from ratings rank, money level, roster size, PLE score, market spend/releases. **No money change.** Negative deltas scaled by difficulty.

7. CPU Rival World

Key file: src/game/cpuRivalLoop.ts

Hidden parallel simulation each Run Show:

1. Build 4-segment card (6 on PLE)
2. Score segments (pop/ring/promo/mom/morale/fatigue + GM style + hash ± 7)
3. Show score = avg + rivalry/title/PLE bonuses + cpuWeeklyScoreModifier (-2 to +4 by difficulty)
4. Resolve CPU titles (change if score ≥ 78 and hash ≥ 58)
5. Roster fallout (simpler injury model than player)
6. Ratings battle standings vs player
7. CPU finance: simplified revenue = $62000 + \text{score} \times 950 + \text{segments} \times 4200 + \text{PLE bonus}$

CPU market on Advance Week: contract aging, renew/release, CPU-to-CPU trades — all hash-gated with difficulty thresholds.

8. Social / IWC Mood

Post generation: generateSocialPosts — up to 7 TV / 8 PLE posts from show outcomes (title scenes, rivalry heat, fatigue concern, overrun, etc.)

IWC mood summary: getIwcMoodSummary — retrospective dominant tone/category from posts for latest week. **Read-only; no mechanical effect on wrestlers.**

9. Season & Calendar

- **12-week season** with PLE cadence on calendar
- Week 12 complete → season review; startNextSeason resets TV stats, calendar, market board; retains contracted roster + money
- Champion passive carry and contract aging continue across seasons

10. Difficulty as Global Pressure Profile

Single source: src/game/difficultyRules.ts

System affected	Easy → Legendary trend
CPU weekly scores	-2 → +4 modifier
CPU draft quality	less value-aware → ruthless
CPU market aggression	higher claim thresholds → lower (more claims)
Player morale penalties	0.75× → 1.30×
Player injury risk	-5 → +7
Office negative deltas	0.75× → 1.30×

System Coupling Map (Where Changes Ripple)

Stats → Segment Score → Show Score → Momentum Gain / Finance Revenue / Social Posts / Rivalry Heat / CPU Ratings Battle

Segment Score → Fatigue Gain → Morale Penalty / Injury Risk

Morale → Segment Score (feedback loop)

Highest-leverage knobs: segment score profiles, momentum tier thresholds, fatigue floors/recovery, injury thresholds (92/97), rivalry freshness decay, finance attendance multipliers, difficulty profile.

Strengths (Current Design)

1. **Clean pre-show / post-show boundary** — no spoiler predictions; consequences are earned
 2. **Fully deterministic player sim** — reproducible for balance testing (statDynamics.test.ts, scoring.test.ts)
 3. **Segment-type differentiation** — distinct weight profiles, fatigue floors, chemistry bonuses
 4. **Rivalry freshness anti-spam** — repeated beats same night heavily taxed
 5. **Difficulty as one profile** — consistent pressure across CPU, market, morale, injury
 6. **Finance v3 decoupled from payroll** — show economics readable per week; market is separate cash events
 7. **Open Challenge** — hidden opponent + readiness modifier creates surprise without breaking determinism
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Review Discussion Areas (Potential Strengthening Targets)

These are **observations for our review**, not prescriptions:

A. Stat economy & pacing

- Momentum gains (+1 to +8 per segment) vs weekly decay (−4 unbooked) may create **starvation spirals** for midcard not booked regularly
- Fatigue recovery (−3/week) vs Match floor (+8) means **2 matches/week** without rest pushes toward injury/morale pressure quickly — is that intended TV realism or too punishing?
- Elite momentum decay (−2 at ≥92) is mild; stars may **plateau high** with regular booking

B. Injury curve

- Thresholds 92/97 with only +0−19 hash variance create a **sharp cliff** — small risk-score changes flip outcomes dramatically
- Physical segment stacking (+12 each, +14 bonus for 2+) heavily punishes long cards with multiple matches for same wrestlers
- CPU injuries use a **separate simpler model** — rival world may feel inconsistent with player roster injury drama

C. Morale as sole "sentiment"

- Morale is both a **segment score input** and a **post-show fallout output** — feedback loop can amplify booking mistakes
- Protected rest (+2, blocks underuse) is powerful but requires social inbox engagement — worth reviewing accessibility

D. Rivalry heat vs freshness

- High scores (+12 heat) vs freshness costs (−3 to −15) — feuds can rise fast but go stale if not spaced across weeks
- Crowd-spark at ≥85 may fire often on good cards — review frequency in full-season playtests

E. Finance feel

- Show P/L can swing widely; market signings are **lumpy prepaid hits** not smoothed across weeks

- \$2M start with ~12–16 picks often leaves **tight reserves** — early market flexibility may be constrained by design
- CPU finance is simplified vs player v3 — ratings battle pressure may not reflect comparable economic stress

F. CPU parity

- CPU match winners/titles use hash gates; player uses pure stats — **intentional asymmetry** but affects ratings battle fairness
- cpuWeeklyScoreModifier is a blunt $\pm 2-4$ — may need per-brand GM style differentiation beyond draft

G. Draft ↔ career balance

- Difficulty doesn't affect starting budget (only \$2M default in UI) — Easy and Legendary start with same purchasing power
- CPU drafts until budget exhausted (not fixed 12) — rival roster depth varies by draft efficiency

H. Missing / deferred mechanics (by design)

- No weekly payroll deduction
- No segment booking cost at booking time (only in post-show expense report)
- No predictive grades or fan reaction pre-show
- Tag team records/team stats not persisted

Key Source Files Index

Domain	Primary file(s)
Show sim	src/game/scoring.ts
Advance week	src/game/advanceWeek.ts
Draft	src/game/openingDraft.ts, top200DraftPool.ts
Finance	src/game/finance.ts, financeCatalog.ts
Market/contracts	src/game/market.ts
CPU rivals	src/game/cpuRivalLoop.ts
Difficulty	src/game/difficultyRules.ts
Title fallout	src/game/titleStatFallout.ts
Stipulations	src/game/stipulationCatalog.ts
Social	src/game/social.ts, src/social/socialReads.ts
Roster reads	src/game/rosterContextReads.ts
Balance tests	src/game/statDynamics.test.ts

Suggested Review Session Agenda

1. **Play-feel targets** — What should a "good" vs "bad" week feel like in momentum/morale/finance?

2. **Pacing** — Is 12-week season long enough for feuds to breathe before freshness decay?
 3. **Injury frequency** — Target majors per season? Current thresholds vs desired drama
 4. **Finance loop** — Is prepaid-only market the long-term model? Reserve pressure after draft?
 5. **CPU fairness** — Ratings battle: should CPU scores correlate tighter with player formula?
 6. **Difficulty** — Should Easy/Legendary also shift starting budget or segment balance constants?
 7. **Priority tuning pass** — Pick 2–3 systems for first bounded balance slice (e.g. fatigue/injury, rivalry freshness, finance attendance)
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Review To-Dos

1. Align on 2–3 priority systems to strengthen first (fatigue/injury, rivalry pacing, finance, CPU parity, etc.)
2. Define target feel: good/bad week outcomes for momentum, morale, finance, and feud arc length
3. After priorities set, run bounded tuning on chosen constants in `scoring.ts` / `finance.ts` / `difficultyRules.ts` with `statDynamics.test.ts` verification